

# Recurrent Flight Training
## 747-400/-8
### Flight Training Lesson Guide
### Trainer Guide

**Revision-**
**10 DEC 25**

## Revisions

Changes to this lesson guide must be approved by the Fleet Captain.

Suggestions or comments shall be directed to the Fleet Captain.

Any questions about the content or use of this manual can be directed to:

Fleet Captain - Atlas Air Training Center or
Standardsreport747@atlasair.com

## Lesson Guide Location

Lesson guides are available on Comply365 under the respective airplane training tab.

Only the most recent lesson guide is listed.

Master lesson guide files are maintained by the Fleet Captain and the Flight Operations Support Group.

## Revision Highlights

Updated 747 lesson training and checking items for Appendix E and F including the following:

### General

- Added Landing Energy Management Briefing
- RFT to be conducted as a freighter flight
- Use of headsets, hand microphones and cockpit speakers not mandatory during RFT

### Briefing Room Items

Passenger procedures reviewed as a briefing room only item.

### Preflight - Taxiing

- Add non-normal engine start(s)
- Taxiing procedures in compliance with instructions issued by the appropriate Traffic Control Authority or by the person conducting the training
- Use of airport diagram (surface movement chart)

**Preflight - Taxiing (Continued)**

- Obtaining appropriate clearance before crossing or entering active runways
- Observation of all surface movement guidance control markings and lighting
- Pretakeoff procedures that include power plant checks, receipt of takeoff clearance and confirmation of aircraft location, and FMS entry (if appropriate), for departure runway prior to crossing hold short line for takeoff
- Initial taxi-out now RVR 1200
- Runway Change on taxi out

**Takeoff**

- Weather / ATIS guidance revised as follows:
  - Initial takeoff now RVR 1200
  - RTO accomplished with RVR 500
- Add takeoff crosswind with gusts
  - Use simulator preset

**Cruise**

- Delete TCAS exercise

**Landing**

- Performance standards revised to reflect stabilized approach criteria
- Delete ILS CAT-II / CAT-III approach
- Add ILS CAT-I autoland (CA & FO)
- Add landing crosswind with gusts
  - Use simulator preset

## Introduction

The Recurrent Flight Training (RFT) program is designed so that two pilots can be trained in pertinent flight crew procedures and maneuvers during a single flight training session in compliance with 121.409(b)(1). The program requires at least 4 hours of training at the pilot controls. Both Pilots are receiving essential “crew concept” training throughout the training session.

The RFT is a single simulator period. Specific training on every event is unnecessary unless it is needed for maintaining pilot proficiency on a particular event. RFT emphasis should be on those events or other maneuvers or procedures not normally encountered during routine line operations. RFT events may be tailored to address observed problem areas during Line Operations.

An RFT segment is deemed incomplete if a Flight Crewmember is unable to perform the required maneuver(s) to proficiency in the allotted training time. An additional training segment must be accomplished to train the airman to proficiency. The airman cannot serve in revenue operations until this training is accomplished. An incomplete RFT may warrant a Proficiency Check.

This flight profile contains procedures and maneuvers that can be evaluated in a Level C/D motion visual simulator.

To the extent practical, realism will be maintained throughout training. The trainer will act as the Company Dispatcher, Ground Maintenance, and ATC. Realistic ATC communications will be conducted at all times, except during specific flight maneuvers (stalls, steep turns etc.).

Use of headsets, hand microphones and cockpit speakers not mandatory during RFT. Flights should be flown in real time except when repositioning or when training specific maneuvers. This does not preclude verbal instructions and comments when appropriate, and position freezing and repositioning to make more efficient use of available time. System failures and faults will be removed after each landing.

The Crew will be expected to use cockpit resources available (autopilot, auto throttle, etc.) unless specifically instructed by the trainer. Judgment, Aircraft Control, Situational Awareness, FMS-CDU Operation, Command Qualities, and Crew Coordination are part of the training.

Each lesson contains multiple non-normal procedures. It is not required for both pilots to perform each procedure. The trainer should attempt to expose the trainees to as many non-normal events as possible.

If the crew pairing is other than CA/FO, an exchange of seats must occur at approximately the midpoint to ensure that training is conducted from the appropriate duty position.

It may be necessary to vary the order of specific items due to circumstances that arise (e.g., unplanned go-around). All items are training events, which means the trainer should provide guidance and input as necessary before and during the training. Any item may be repeated as time permits to achieve proficiency.

Repositioning may be used to ensure completion of all RFT elements.

## Performance Standards

Trainees are individually responsible for familiarity and correct execution of Company and FAA procedures. Your trainer will insist you learn and demonstrate the correct procedures. The Evaluation Criteria Section contains standards for maneuvers and events in the FFS program. These standards conform to FAA Order 8900.1 Airline Transport Pilot Certificates, Flight Test Events in Airplanes.

<table>
<tr><th>TAXI-OUT</th><th>TAKEOFF</th></tr>
<tr><td>Maintain taxiway centerline, <15kts speed in congested areas, and <30kts in non-congested areas.</td><td>• Standard Callouts<br>• Rotate at VR<br>• Rotation Rate 2.5°–3.0°/s<br>• Maintain RW Centerline</td></tr>
<tr><th colspan="2">CLIMB/CRUISE/DESCENT/HOLDING</th></tr>
<tr><td colspan="2">• Airspeed..... ± 10kts<br>• Altitude ..... + 100 / -100 ft.<br>• Heading..... ± 10°<br>• Course ..... ± 5°<br><br>From rotation to low altitude level-off, maintain:<br>• Positive rate of climb</td></tr>
<tr><th>APPROACH (PRECISION)</th><th>APPROACH (NON-PRECISION)</th></tr>
<tr><td>From FAF / OM to DH, maintain:<br>• Airspeed..... +5 / -5 kts<br>• Localizer..... 1/2 dot<br>• Glideslope ..... 1/2 dot</td><td>From FAF to MDA(H), maintain:<br>• Airspeed.....+5 / -5kts<br>• Less than half-scale deflection of the Course Deviation Indicator (CDI) or +/- 5° in the case of the RMI or bearing pointer.<br>• VNAV - Altitude +75/-75 ft from vertical track<br><br>From MDA(H) to MAP, maintain:<br>• Altitude.....+50/ - 0 ft<br>• Less than half-scale deflection of the CDI or +/- 5 degrees in the case of the RMI or bearing pointer.</td></tr>
<tr><th colspan="2">BELOW 1000’ AGL</th></tr>
<tr><td colspan="2">Maintain “Stabilized Approach” criteria</td></tr>
</table>

**Scheduled briefing and simulator times are as follows: (Hr + Min)**

Pre-Briefing Times

RFT 2+00

De-Briefing Times

RFT 1+00

Simulator Times

RFT 4+00

## Objective

To train the skills and knowledge required to accomplish the maneuvers, procedures, and duties as captain or first officer.

## Training

- Portable Electronic Flight Bag (pEFB)
- Glare-shield Checklists
- Flight Data / Setup Sheet which includes:
  - Setup Sheet
  - DDG
  - Fuel Data
  - Weather
  - Weight & Balance Data
  - Clearance
  - TLR / FDP

## Weather / ATIS Guidance

- Taxi / Takeoff
  - Initial Taxi RVR 1200
  - RTO accomplished with RVR 500
  - Issue a runway change on taxi out.
- ILS approaches
  - The Runway Visual Range (RVR) should be set to the minimum value specified for the approach.
  - The ceiling should be set to a value to allow visual acquisition of the runway approximately 50 to100 feet above the Decision Altitude (DA).
- Non-ILS approaches
  - A visibility value of approximately ¼ mile greater than the published minimums value.
  - If LNAV / VNAV is to be used with a published DA, the visibility should be set to 2 miles.
  - The ceiling should be set to a value to allow visual acquisition of the runway approximately 50 to100 feet above the Decision Altitude (DA).
- Landings
- Visual approaches shall be set as CAVOK
- Engine failure after V1 and before V2 shall be conducted at CAT-I ILS minimums with a crosswind of 12kts

ATIS may be in printed, electronic format or verbalized.

If verbalized, ATIS should be provided with proper ATC phraseology.

Example:
***"Kennedy International Information*** *"Alpha" XXxxz, Wind 220/10, Visibility 1/2sm FG Sky condition Overcast 200, Temperature 08, Dew Point 06, Altimeter 29.92, Departing Runways 31L. On initial contact advise you have information Alpha".*

## Briefing Room Items

- Review paperwork and crewmember's documentation
- Review students Comply365 for current manuals, Jepp FD Pro, and all compliance documents are completed.
- **Landing Energy Management**

**Core Principle**
Landing Energy Management is an ADM process – not a checklist. FDP provides cues, not directives. The pilot decides how to manage each landing using available tools.

**What Drives Brake Energy**
- Small speed changes = large brake-energy changes.
- Mass contributes linearly; speed contributes exponentially.
- Most brake energy is absorbed early in rollout when groundspeed is highest.

**FDP – Use as a Decision Tool**
- Enter expected weight, winds, and runway state.
- If conditions are uncertain, bias toward a worst-credible scenario.
- Use FDP to compare configurations (flaps, reverse, autobrake).
- Re-run FDP when conditions meaningfully change.
- FDP predicts brake temps and stopping distance – these are cues, not limits.

**Aeronautical Decision Making (ADM)**
FDP Sequence (typical)

Flaps → Reverse Thrust → Autobrake → Runway

- Flaps 25 = baseline; Flaps 30 is often the first and most economical mitigation.
- Reverse thrust is most effective at high speed.
- Select the lowest autobrake setting that meets runway condition needs.
- Execute the rollout you planned; avoid early turnoffs.

**747-Specific Note**
- Reverse thrust on the 747 carries a unique FOD risk: engines 2/3 can lift debris into engines 1/4.
- Crews transitioning from twins must understand the geometry-driven FOD hazard unique to the 747.
- Reverse remains a pilot tool when conditions warrant it.
- Crews apply reverse thrust based on ADM and FDP results – not habit.

- **Passenger Operations Review (Briefing room only)**
  - Passenger Operations Briefing Guide
  - Purser – Flight Attendant Briefing
  - Flight Deck Door Procedures
    - General
    - Flight Crewmember Leaving the Flight Deck (Pax Ops)
    - Cabin Crew Entering the Flight Deck
    - Flight Deck Door Failure
    - Keypad and Dead Bolt
  - Communications:
    - Normal Crew Communications
    - Non-Normal Crew Communications
    - Normal and Non-Normal Communication Glare-shield Card
    - "FLIGHT DECK REPORT" – Call the Purser to the flight deck for TEST checklist briefing
    - "CABIN CREW TO STATIONS" – If time does not allow a flight deck report: Call the cabin crew to their stations for TEST checklist briefing.

## Briefing Room Items (Continued)

- **FDP / TLR Review**
  - How to read a TLR
    - What altitude to start turn on a “Simple” engine failure procedure (EFP)? 1000’ AGL
    - Special and Complex EFP per FDP/TLR if applicable.
  - Performance via ACARS (FDP)
- **Supplementary Procedures**
  - Manual Thrust Lever Takeoff
- **Non-Normal Procedures**
  - Fire Aft Cargo and Evacuation
- **Approach and Landing**
  - Stabilized approach criteria
  - Crosswind Limitations
  - Touchdown Zone Marking
  - Use of Reverse Thrust
  - Non-ILS Approach (Using V/S)
  - Non-ILS Approach (Using VNAV)
- **Rejected Takeoff**
  - Decision Criteria
  - Responsibility and Control
  - CA / FO Responsibilities
  - Non Normal Communication
- **Rejected Takeoff (FO perform PIC duties)**
  - **747 - After the aircraft is aligned with the runway, the FO will:**
    - Release the brakes if previously set
    - Smoothly advance thrust levers to approximately GE 70%N1, GEnx 45%N1, and allow engines to stabilize
    - Push TO/GA switch to advance the thrust levers to takeoff thrust, or manually advance the thrust levers to takeoff thrust
    - Keep thumb on the autothrottle disconnect switch until 80 knots
    - Keep hand on the thrust levers until the V1 call and be ready to abort prior to V1 if necessary, as per Atlas SOPs.

## Briefing Room Items (Continued)

.

***Note:*** ***With the exception of this training / checking maneuver, the FO will not normally be expected to perform the CA rejected takeoff duties during actual line flying.***

- **Takeoff with engine failure after V1 and before V2**
  - Aircraft control
  - A/P Engagement
  - ATC Communication
  - Engine Failure Procedure (EFP)
  - Immediate Action Items
  - Flap Retraction and Thrust Management
  - Non Normal Checklist (NNC)
  - Engine Inoperative Climb (VNAV)
- **One Engine Inoperative ILS Approach**
  - Profile
  - Use of Rudder Trim
- **Rejected Landing**
  - One Engine Inoperative
  - Failure of Second Engine on Initial Climb
- **Visual approach and landing with 50% loss of thrust**
  - Without external glide path (G/S) or PAPI/VASI
  - Both engine failures must be on the same side

## Simulator Lesson Summary

The RFT lesson guide is to be used as the lesson plan for Recurrent Flight Training (RFT)

- Lesson to be conducted as a Freighter flight
- FDP and TLR Review
- Begin period at the gate with EXT electrical power established
- Normal crew station preflight
- If crew pairing consists of CA/CA, each CA is required to perform his/her RFT from the left seat. The same applies to the right seat if crew pairing is FO/FO
- For CA/CA and FO/FO pairings, after a seat swap at half-time, execute for the second PF, the first half of the lesson plan again, beginning from the Takeoff section
- Normal and Non-normal engine starts
- Low visibility instrument conditions taxi RVR 1200
    - Runway incursion avoidance awareness
    - TCAS Use / NAV display range / Orientation
- **Takeoff**
    - **Normal takeoff – Crosswind**
    - **At least one takeoff with crosswind and gusts**
    - **Rejected Takeoff (CA) and (FO)**
        - Both CA and FO shall conduct PIC RTO duties. Reference briefing items for information on how to conduct the FO PIC RTO
        - Lowest Authorized Visibility
        - Set the malfunction to occur as close as possible to V1, yet allowing the PIC enough time to perceive and respond to the problem prior to reaching V1
        - The malfunction should be one that clearly and unequivocally requires rejection of the takeoff
        - On occasion the malfunction selected should lead to an evacuation of the aircraft

- **Takeoff with engine failure occurring after V1 and before V2**
  - The engine failure should occur after V1 and before V2
  - The engine failure must be an outboard engine
  - The engine failure may be on the upwind or downwind engine

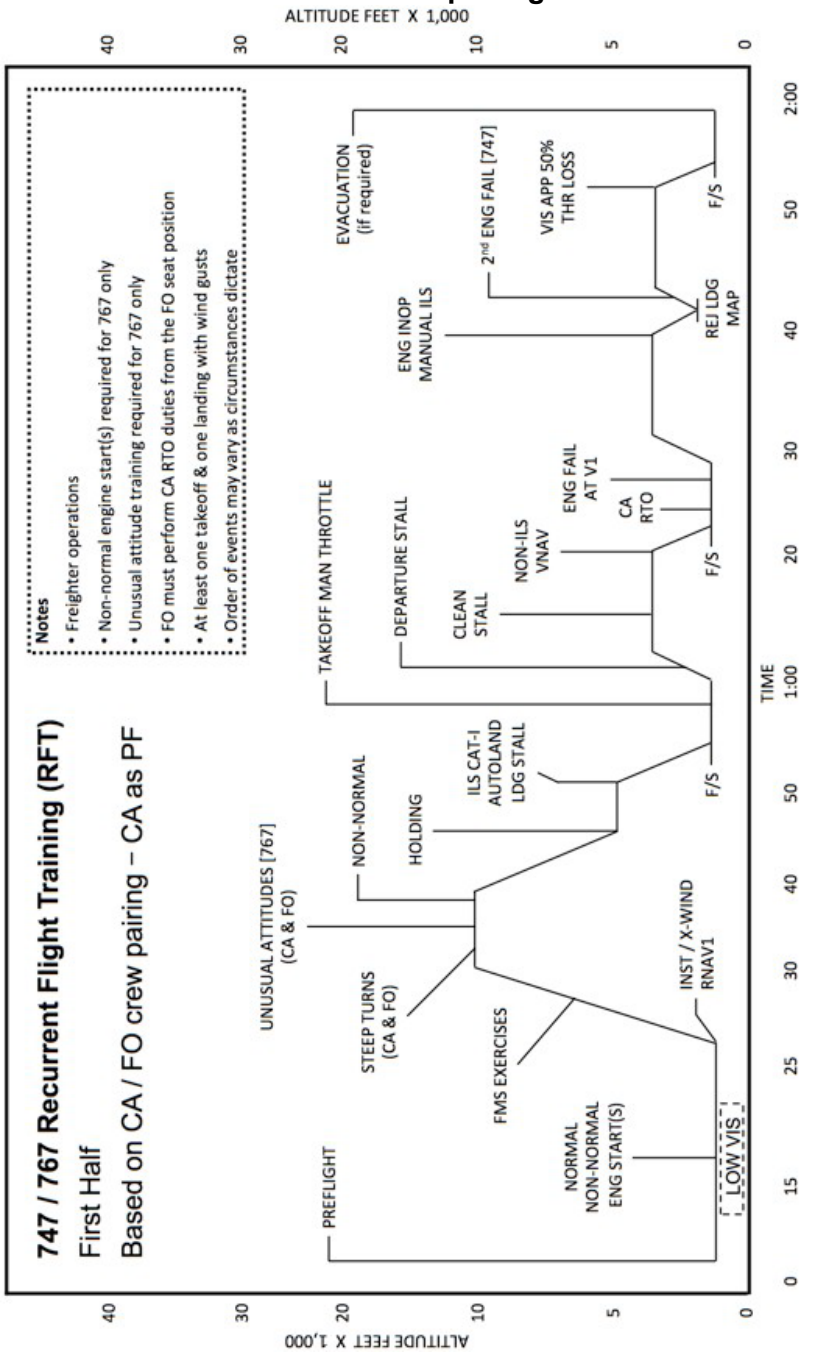
- **RNAV departure procedure**
- **LNAV / VNAV Exercises**
  - Exercises may be varied or added on to by the instructor. Use position freeze as necessary to avoid traveling from the area.
  - Training goal is FMC programming proficiency. Once CDU entries are correctly made, it is not necessary for the crewmember intercept the radial/course.
- **Air-work (Steep Turns and Stalls)**
  - **Steep Turns**
    - Two 180 deg steep turns
    - Accomplished outside of ATC environment
  - **Stalls**
    - Clean
    - Landing
    - Takeoff
    - Incorporated into their respective phase of flight
    - Reference Stall Training Supplement Located on Comply365

Non-Normal procedures

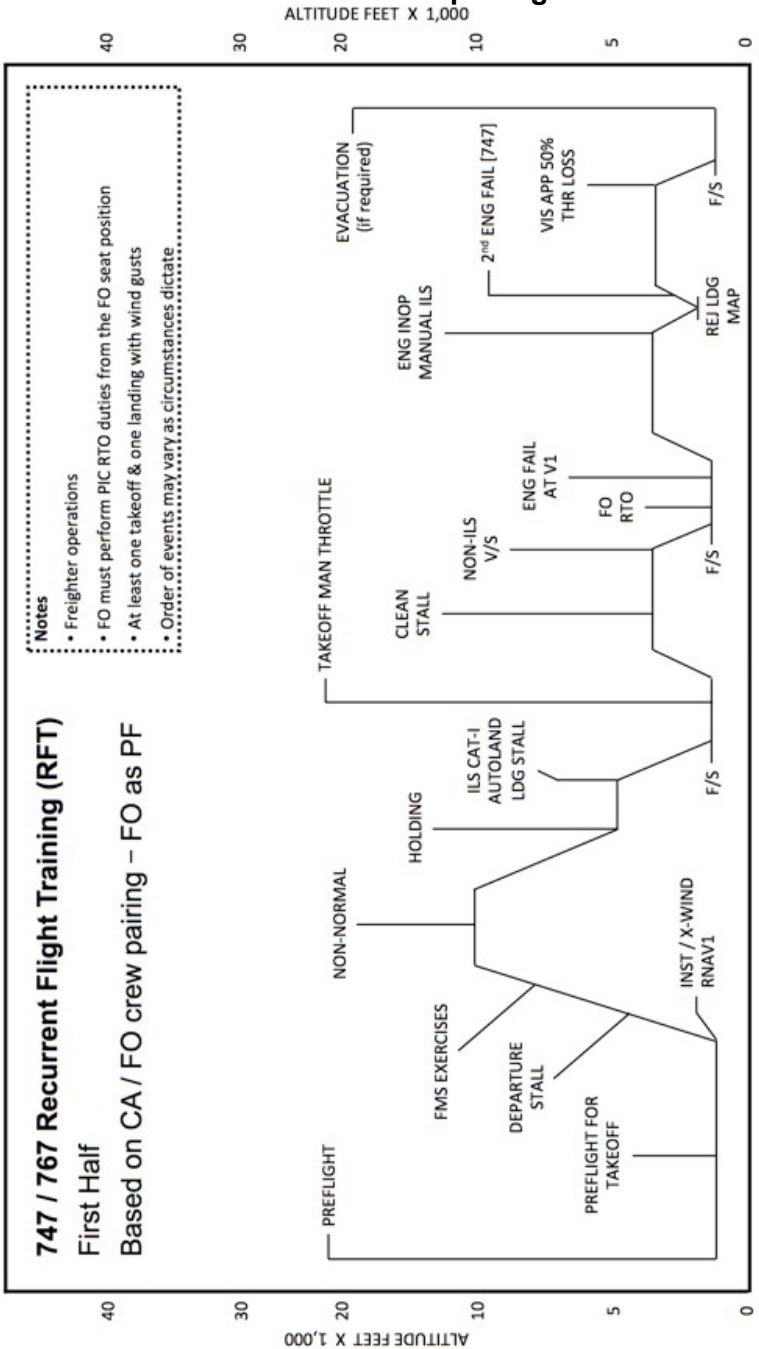
- **Holding**
  - Should be incorporated in a realistic manner and include entering, maintaining, and leaving holding patterns

- **ILS Approach**
  - ILS CAT-I all-engine Autoland approach CA and FO
  - ILS CAT-I approach one engine inoperative, manually flown to a rejected landing or missed approach
    - For training / testing purposes the one engine inoperative manually flown approach must be flown with A/P and A/T off. Manual flight should commence on base leg
    - A crosswind component of 12 knots should be used to evaluate the crewmembers ability to track the localizer course
    - The approach should terminate with a rejected landing / missed approach procedure
- **Non-ILS Approach**
  - One non-ILS approach for each crewmember
  - VNAV as pitch mode 1st Half
  - V/S as pitch mode 2nd Half
  - Each Non-ILS Approach should begin from a point sufficiently above the initial approach fix altitude to allow the crewmember to demonstrate energy management, vertical situational awareness and VNAV / V/S control
- **Visual approach and landing with 50% loss of thrust**
  - Both CA and FO shall perform a visual approach and Landing with 50% loss of thrust.
  - CAVOK
  - This approach shall be flown without external glide path (G/S) or PAPI/VASI
  - Both engine failures must be on the same side
- **Landings**
  - A total of three manually flown landings must be accomplished
  - At least one landing with crosswind and gusts – May be combined with any other landing event
  - One rejected landing may be counted as a landing if the aircraft was clearly in position to land in the touchdown zone without excessive maneuvering
  - One all-engine ILS CAT-I Autoland as PF (CA) and (FO)
- **Rejected Landing**

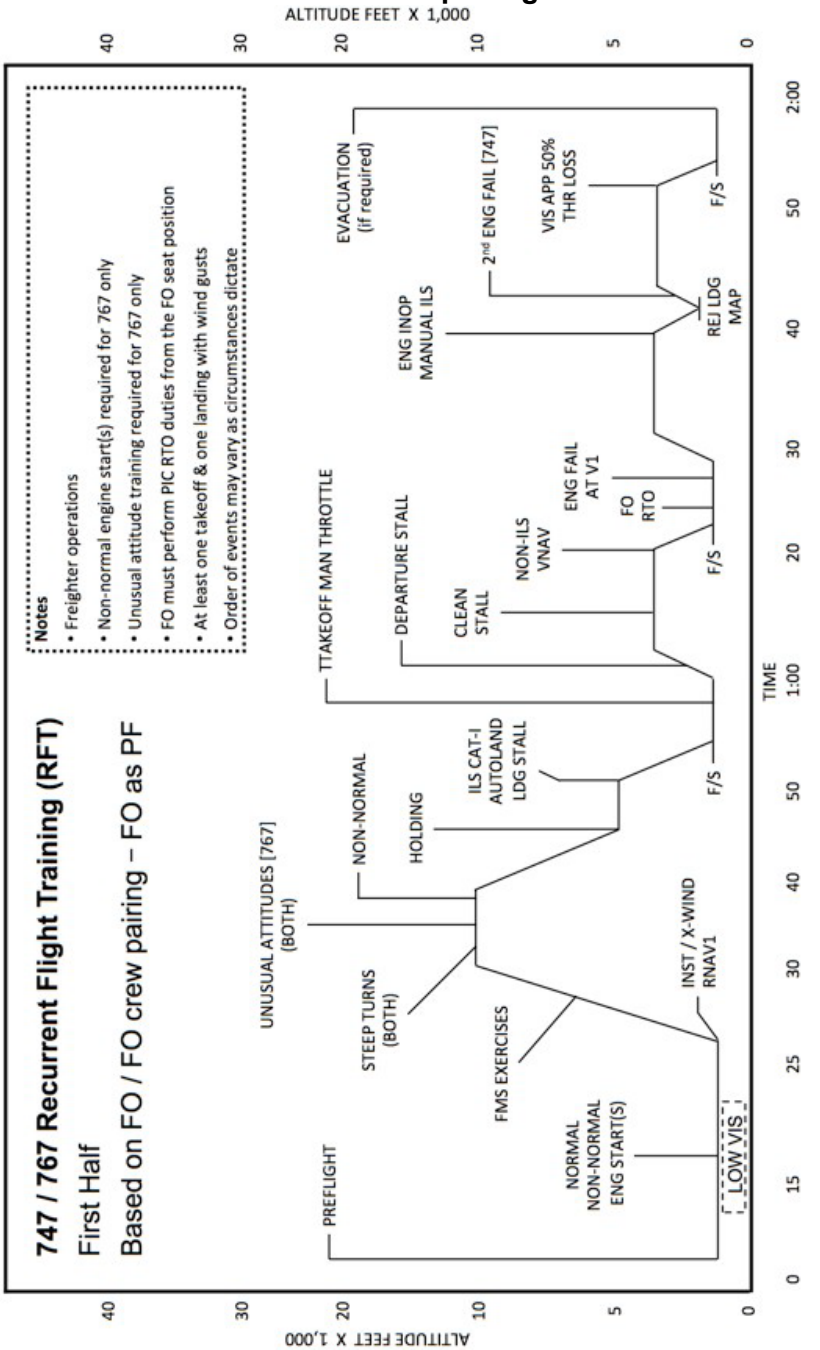
## RFT Profile 1st Half – CA / FO crew pairing CA as PF



## RFT Profile 2nd Half - CA / FO crew pairing FO as PF



## RFT Profile 1st Half - FO / FO crew pairing FO as PF



## RFT Profile 2nd Half - FO / FO crew pairing FO as PF

